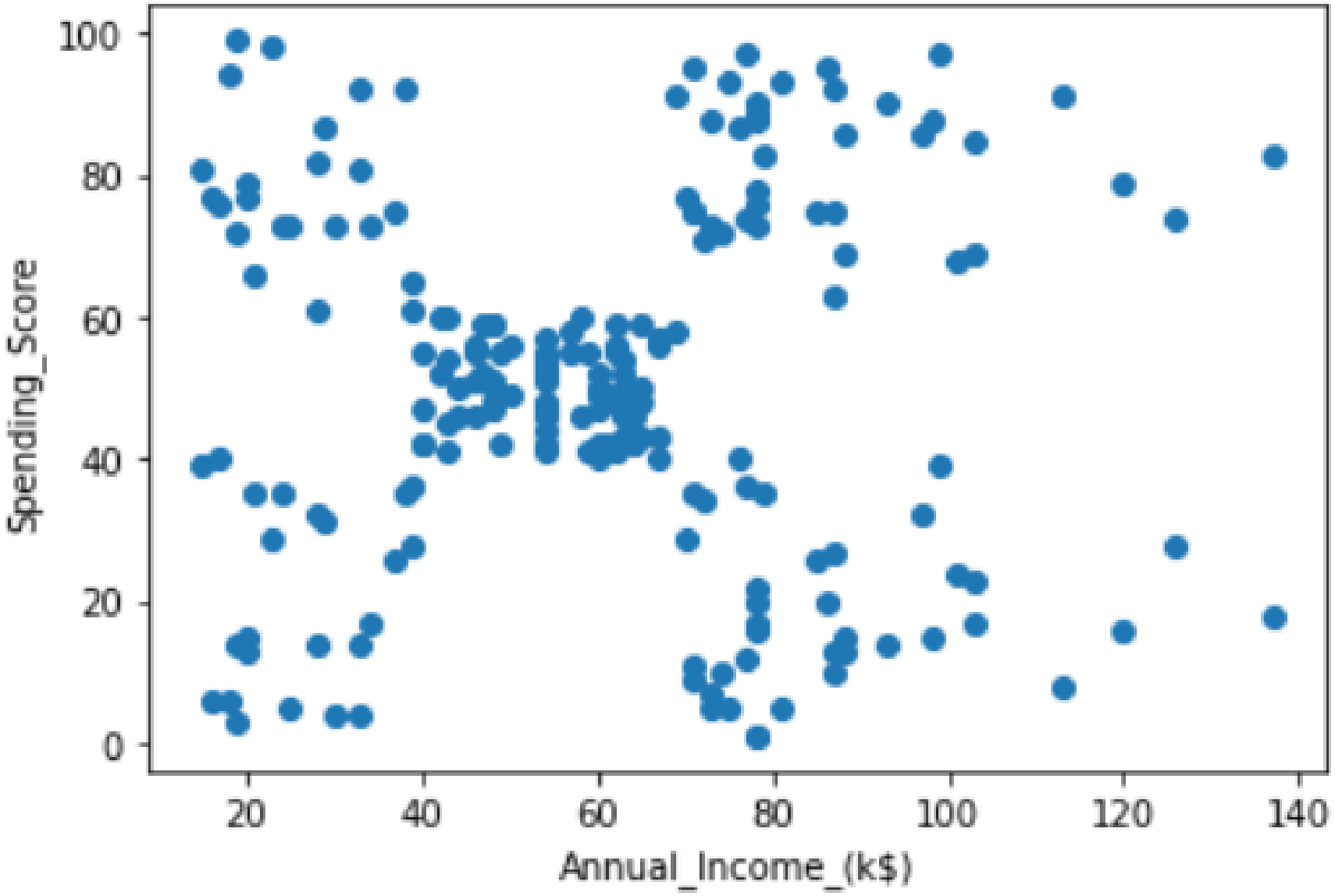
Unsupervised Learning

Clustering methods: **K-means**

Problem 1: Suppose you are working with unlabelled data containing customer information and want to find patterns and similarities in the dataset.

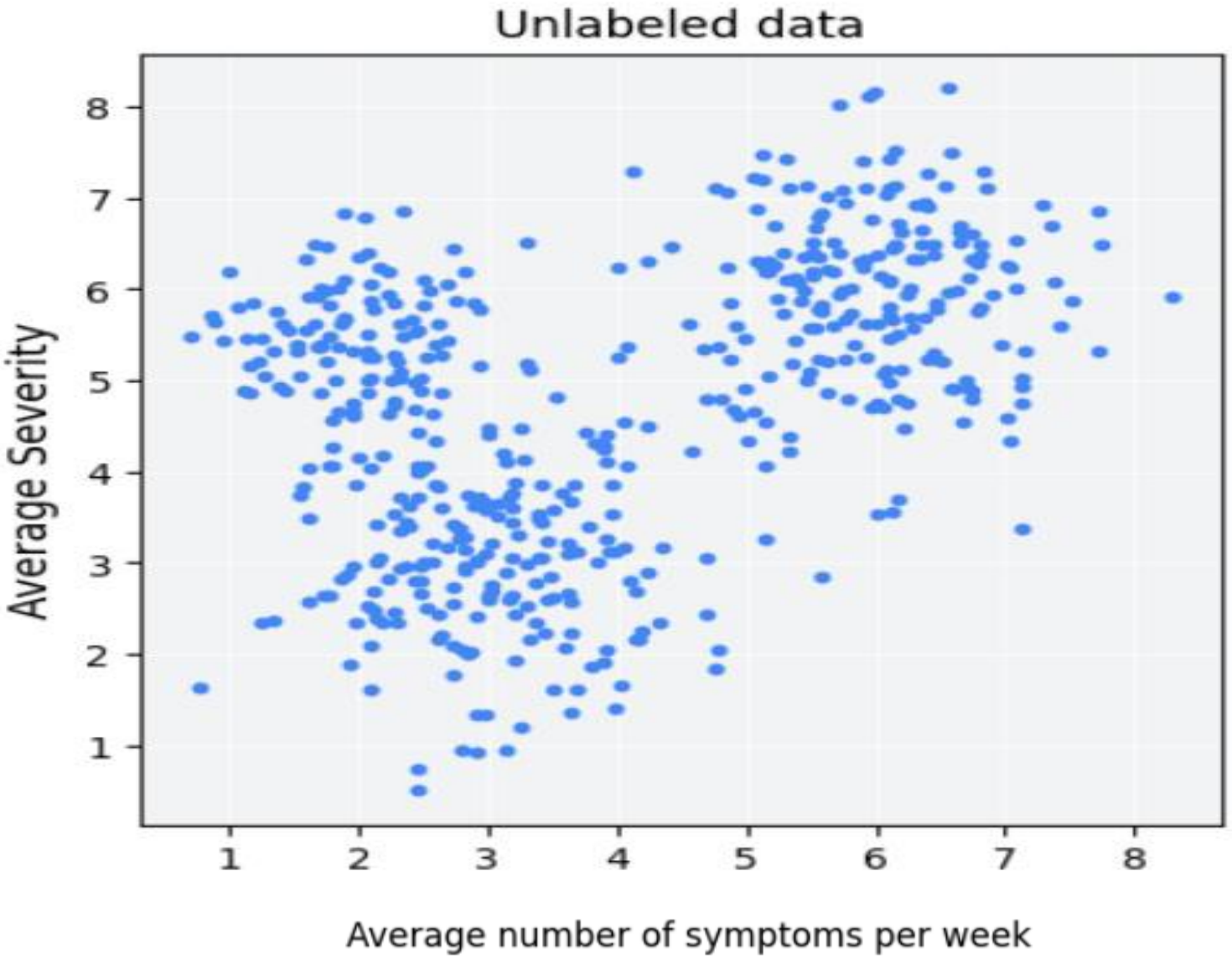
Customer_ID رقم الزبون	Genre الجنس	Age السن	Annual_Income الدخل السنوي	Spending_Score درجة الإنفاق
1	Male	19	15	39
2	Male	21	15	81
3	Female	20	16	6
4	Female	23	16	77
5	Female	31	17	40
6	Female	22	17	76
7	Female	35	18	6
8	Female	23	18	94
9	Male	64	19	3
10	Female	30	19	72
.				
.				
.				
200	Male	30	137	83

Dataset



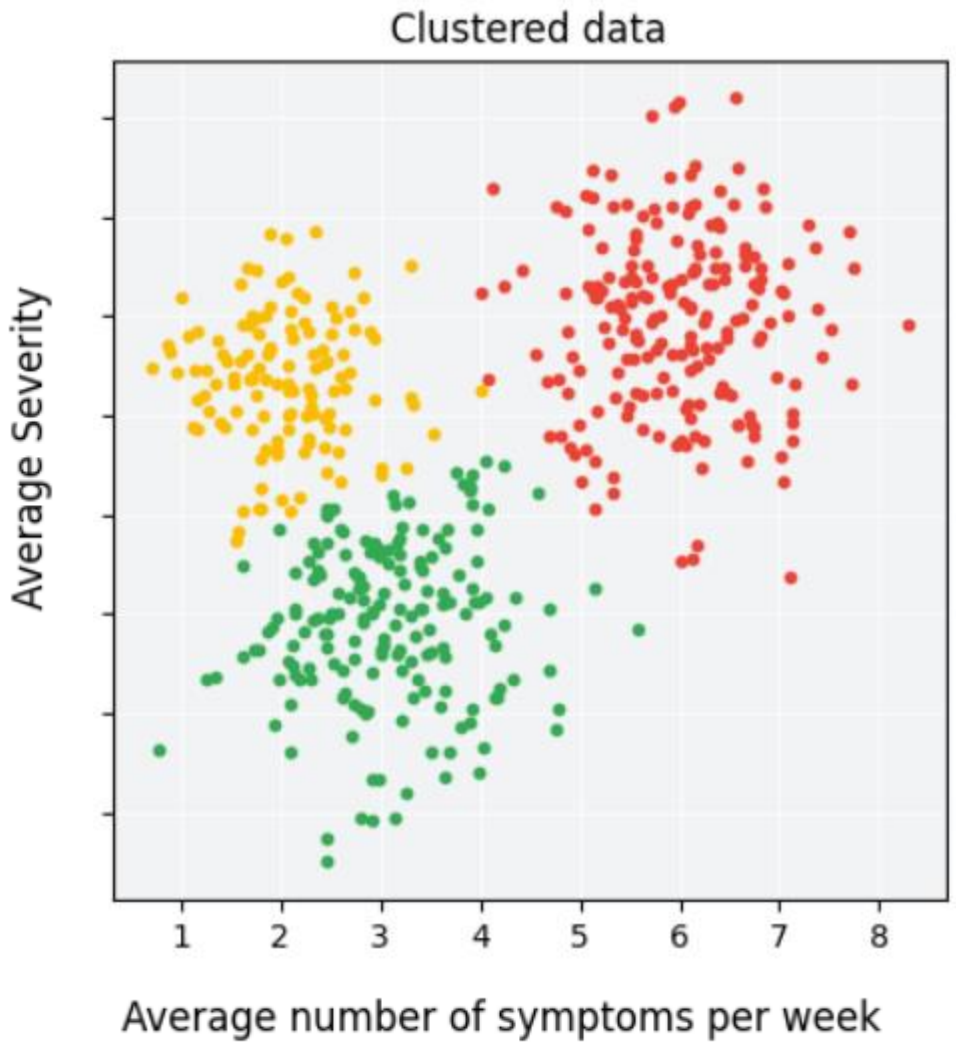
Scatter plot presentation based **Annual_Income_(k$) and Spending_Score**

Problem 2: Example of a dataset that includes patient information from a healthcare system. The dataset is complex and includes both categorical and numeric features. You want to find patterns and similarities in the dataset. How might you approach this task?



Scatter plot presentation based **Average number of symptoms and Average Serverity**

Solution: Researchers can use clustering analysis to group patients with similar treatment responses into clusters. The figure below demonstrates one possible grouping of simulated data into three clusters.



What is clustering?

Clustering is an unsupervised machine learning algorithm that organizes and classifies different objects, data points, or observations into **groups** or **clusters** based on **similarities** or patterns.

Clustering models aim to group data into distinct "clusters" or groups.

Interclass and **Intraclass** :

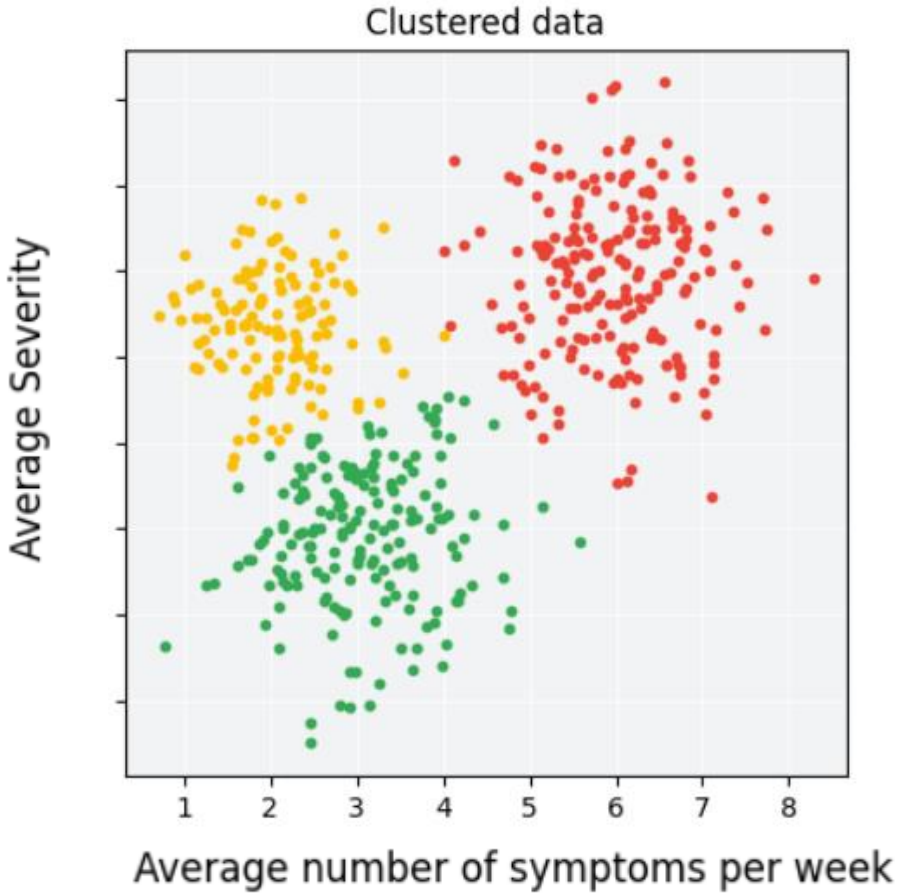
The intra-class cluster shows the distance between data points and the cluster centre; meanwhile, the inter-class cluster shows the distance between data points within one cluster and the other data points in the other cluster.

High intra-class similarity:

cohesive within cluster.البيانات متماسكة داخل المجموعة

Low inter-class similarity:

distinctive between clusters. البيانات مميزة بين المجموعات



What is K-Means Clustering?

K-means clustering is a popular unsupervised machine learning algorithm used for partitioning a dataset into a pre-defined number of clusters. The goal is to group similar data points together and to identify the K number of groups in the dataset.

"*k*" represents the number of groups or clusters we want to classify our items into.

It is an iterative process of assigning each data point to the groups and slowly data points get clustered based on **similar features**.

The objective is to minimize the sum of distances between the data points and the cluster centroid, to identify the correct group each data point should belong to.

There are several distance metrics available that can be used to calculate the distance.

How does K-means work?

1. **Choosing the number of clusters**: The first step is to define the K number of clusters in which we will group the data.

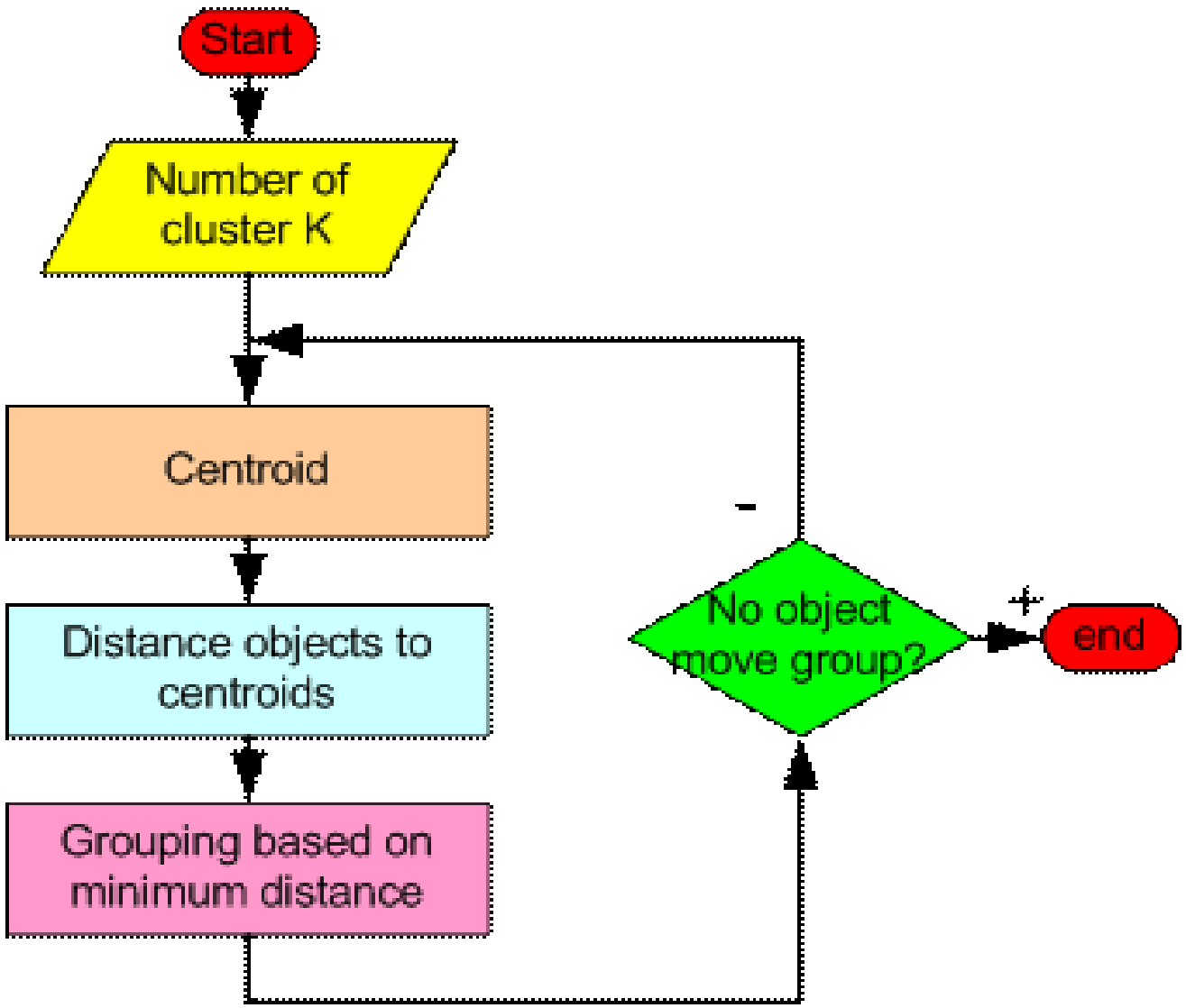
2. **Initializing centroids:** A Centroid is the centre of a cluster, but initially, the exact centre of data points will be unknown, so we select random data points and define them as centroids for each cluster. We will initialize K centroids in the dataset.
3. **Assign data points to the nearest cluster:** In this step, we will first calculate the distance between data point X and centroid C using the **Euclidean Distance metric**. Then choose the cluster for data points where the distance between the data point and the centroid is **minimum**.

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

4. **Re-initialize centroids:** Next, we will re-initialize the centroids by calculating the average of all data points of that cluster.

$$Ci = \frac{1}{|Ni|} \sum xi$$

5. **Repeat steps 3 and 4:** We will keep repeating steps 3 and 4 until we have optimal centroids and the assignments of data points to correct clusters are not changing anymore.


Start
Number of cluster K
Centroid
Distance objects to centroids
Grouping based on minimum distance
No object move group?
-
+
end

Example :

Instance	X	Y
1	185	72
2	170	56
3	168	60
4	179	68
5	182	72
6	188	77

Centroid for first cluster c1 = (185, 72)
Centroid for second cluster c2 = (170, 56)

Iteration 1: Now calculating similarity by using *Euclidean distance* measure as:

d(c1, 3) = $\sqrt{(185-168)^2+(72-60)^2} = \sqrt{(17)^2+(12)^2} = \sqrt{289+144} = \sqrt{433}$

d(c2, 3) = $\sqrt{(170-168)^2+(56-60)^2} = \sqrt{(2)^2+(-4)^2} = \sqrt{4+16} = \sqrt{20}$

Here, d(c2, 3) < d(c1, 3)

So, data point 3 belongs to c2.

d(c1, 4) = $\sqrt{(185-179)^2+(72-68)^2} = \sqrt{(6)^2+(4)^2} = \sqrt{36+16} = \sqrt{52}$

d(c2, 4) = $\sqrt{(170-179)^2+(56-68)^2} = \sqrt{(-9)^2+(-12)^2} = \sqrt{81+144} = \sqrt{225}$

Here, d(c1, 4) < d(c2, 4)

So, data point 4 belongs to c1.

d(c1, 5) = $\sqrt{(185-182)^2+(72-72)^2} = \sqrt{(3)^2+(0)^2} = \sqrt{9}$

d(c2, 5) = $\sqrt{(170-182)^2+(56-72)^2} = \sqrt{(-12)^2+(-16)^2} = \sqrt{144+256} = \sqrt{400}$

Here, d(c1, 5) < d(c2, 5)

So, data point 5 belongs to c1.

d(c1, 6) = $\sqrt{(185-188)^2+(72-77)^2} = \sqrt{(-3)^2+(-5)^2} = \sqrt{9+25} = \sqrt{34}$

d(c2, 6) = $\sqrt{(170-188)^2+(56-77)^2} = \sqrt{(-18)^2+(-21)^2} = \sqrt{324+441} = \sqrt{765}$

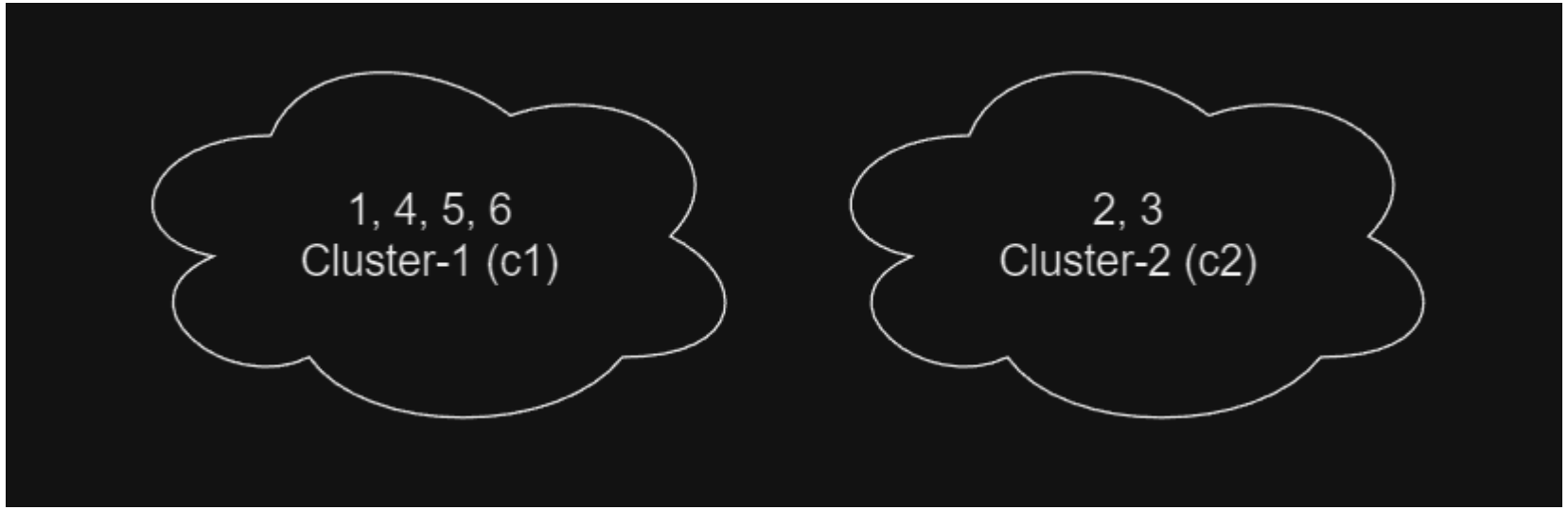
Here, d(c1, 6) < d(c2, 6)

So, data point 6 belongs to c1.

Representing above information in tabular form:

Instance	X	Y	Distance(C1)	Distance(C2)	Cluster
1	185	72			c1
2	170	56			c2
3	168	60	√433	√20	c2
4	179	68	√52	√225	c1
5	182	72	√9	√400	c1
6	188	77	√34	√765	c1

The resulting cluster after first iteration is:



Iteration 2: Now calculating centroid for each cluster:

$$Centroid\ for\ first\ cluster\ c1 = \left(\frac{185+179+182+188}{4}, \frac{72+68+72+77}{4}\right) = (183.5, 72.25)$$

$$Centroid\ for\ second\ cluster\ c2 = \left(\frac{170+168}{2}, \frac{56+60}{2}\right) = (169, 58)$$

Now, again calculating similarity:

$$d(c1,1) = \sqrt{(183.5-185)^2 + (72.25-72)^2} = 1.5207$$
$$d(c2,1) = \sqrt{(169-185)^2 + (58-72)^2} = 21.2603$$

Here, d (c1, 1) < d (c2, 1)
So, data point 1 belongs to c1.

$$d(c1,2) = \sqrt{(183.5-170)^2 + (72.25-56)^2} = 21.1261$$
$$d(c2,2) = \sqrt{(169-170)^2 + (58-56)^2} = 2.2361$$

Here, d (c2, 2) < d (c1, 2)
So, data point 2 belongs to c2.

$$d(c1,3) = \sqrt{(183.5-168)^2 + (72.25-60)^2} = 19.7563$$
$$d(c2,3) = \sqrt{(169-168)^2 + (58-60)^2} = 2.2361$$

Here, d (c2, 3) < d (c1, 3)
So, data point 3 belongs to c2.

$$d(c1,4) = \sqrt{(183.5-179)^2 + (72.25-68)^2} = 6.1897$$
$$d(c2,4) = \sqrt{(169-179)^2 + (58-68)^2} = 14.1421$$

Here, d (c1, 4) < d (c2, 4)
So, data point 4 belongs to c1.

$$d(c1,5) = \sqrt{(183.5-182)^2 + (72.25-72)^2} = 1.5207$$
$$d(c2,5) = \sqrt{(169-182)^2 + (58-72)^2} = 19.1050$$

Here, d (c1, 5) < d (c2, 5)
So, data point 5 belongs to c1.

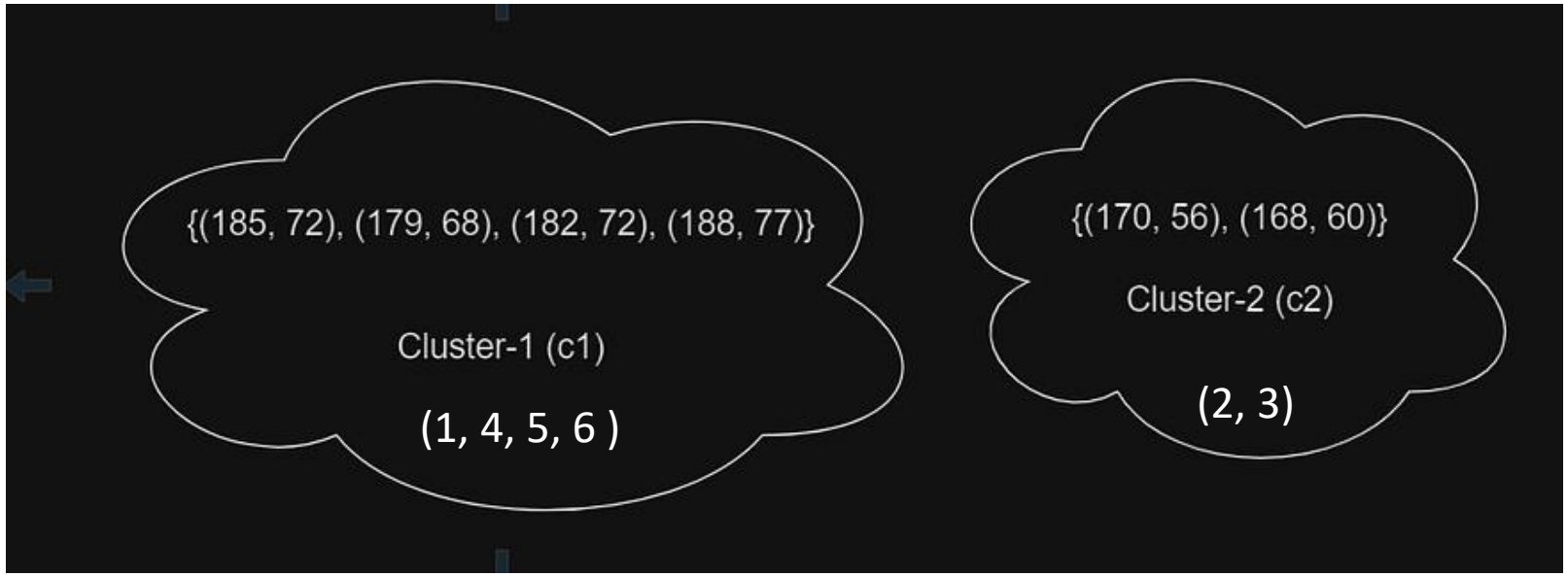
$$d(c1,6) = \sqrt{(183.5-188)^2 + (72.25-77)^2} = 6.5431$$
$$d(c2,6) = \sqrt{(169-188)^2 + (58-77)^2} = 26.8701$$

Here, d (c1, 6) < d (c2, 6)
So, data point 6 belongs to c1.

Representing above information in tabular form.

Instance	X	Y	Distance(C1)	Distance(C2)	Cluster
1	185	72	1.5207	21.2603	c1
2	170	56	21.1261	2.2361	c2
3	168	60	19.7563	2.2361	c2
4	179	68	6.1897	14.1421	c1
5	182	72	1.5207	19.105	c1
6	188	77	6.5431	26.8701	c1

The resulting cluster after second iteration is:



Advantages of the K-means clustering algorithm:

- Relatively easy to understand and implement.
- Scalable to large datasets.
- Better computation cost.
- Easily warm start the assignments and positions of the centroids.

Disadvantages of K-means clustering algorithm:

- Choosing K manually and being dependent on the initial values
- Lacks consistent results for different values of K
- Always tries to find circular clusters
- Curse of dimensionality, K is ineffective when the number of dimensions increases

How to choose the value of "K number of clusters“

The performance of the K-means clustering algorithm depends upon highly efficient clusters that it forms. But choosing the optimal number of clusters is a big task. There are some different ways to find the optimal number of clusters.

1. Elbow method
2. Selhouette method
3. ……

How to select the optimal number of clusters (K) for the dataset.

Here we are discussing the most appropriate method to find the number of clusters or value of K. The method is given below:

1. Elbow method:

Step 1: Apply the k-means clustering algorithm and from the clusters for different values of k.
for example, k=[1,2,3,4,5,6,7,8,9,10]

Step 2: For each k calculate the **Within-Cluster Sum of Squares** (**WCSS**).

$$\text{WCSS}= \sum_{C_k}^{C_m}\left(\sum_{d_i}^{d_n} distance(d_i, C_i)^2\right)$$

Where, **C** is the cluster centroids and **d** is the data point in each Cluster.

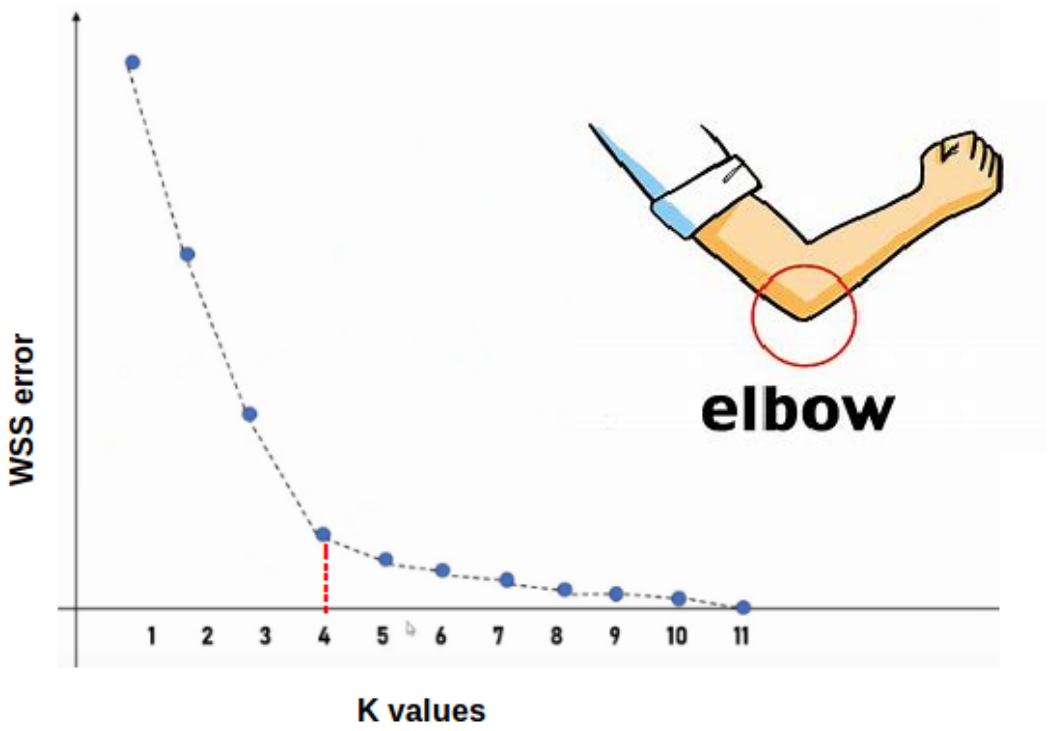
For example, if k=3.

$$\text{WCSS}=\sum_{d\ in\ C_1} distance(d_i, C_1)^2 + \sum_{d\ in\ C_2} distance(d_i, C_2)^2 + \sum_{d\ in\ C_3} distance(d_i, C_3)^2$$

Step 3 : Plot curve of **WCSS** according to the number of clusters.
Step 4 : The location of bend in the plot is generally considered an indicator of the approximate number of clusters.
or a point of the plot looks like an arm, then that point is considered as the best value of K.

- When the value of k is 1, the WCSS will be high. As the value of k increases, the WCSS value will decrease.

Elbow method



- By using the Elbow method, we can select k value, i.e k=4, after 4, **WCSS** is constantly decreasing

Practical exercise:

We will be working on the Mall Visitors dataset to create customer segmentation to define a marketing strategy. it summarises the spendings of around 200 mall visitors.

1. Load the data and check for any missing values.

2. Draw the scatterplot by using the **Annual Income** and **Spending Score** features to find the clusters in the data.

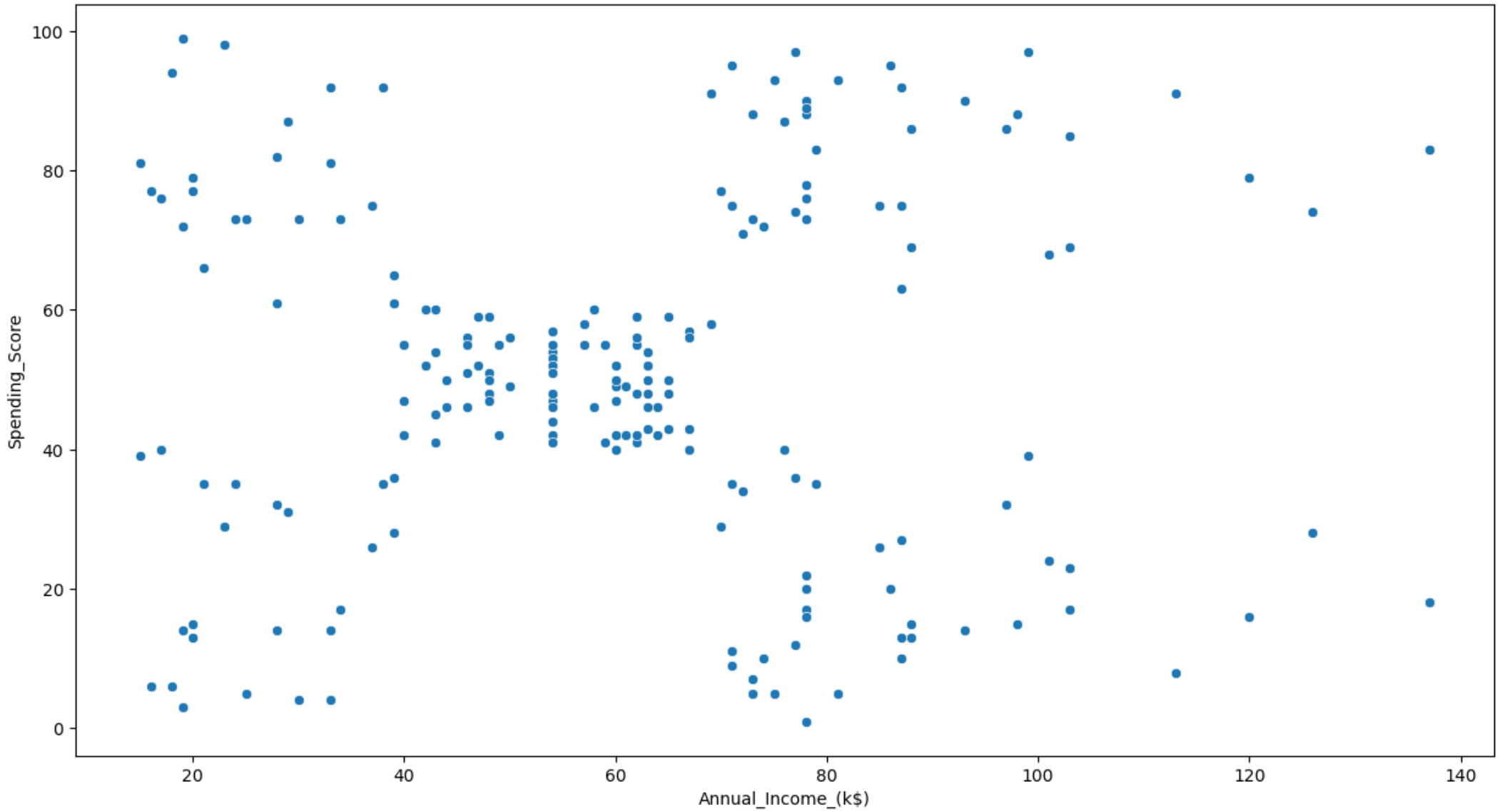
3. Initialize the centroids randomly for k=3.

4. Select the optimal number of clusters (K) for the dataset.

5. Calculate the distance between each centroid and data point, find the K clusters and assign the data points to a significant cluster. This process will continue until the difference between previously defined centroids and current centroids is zero

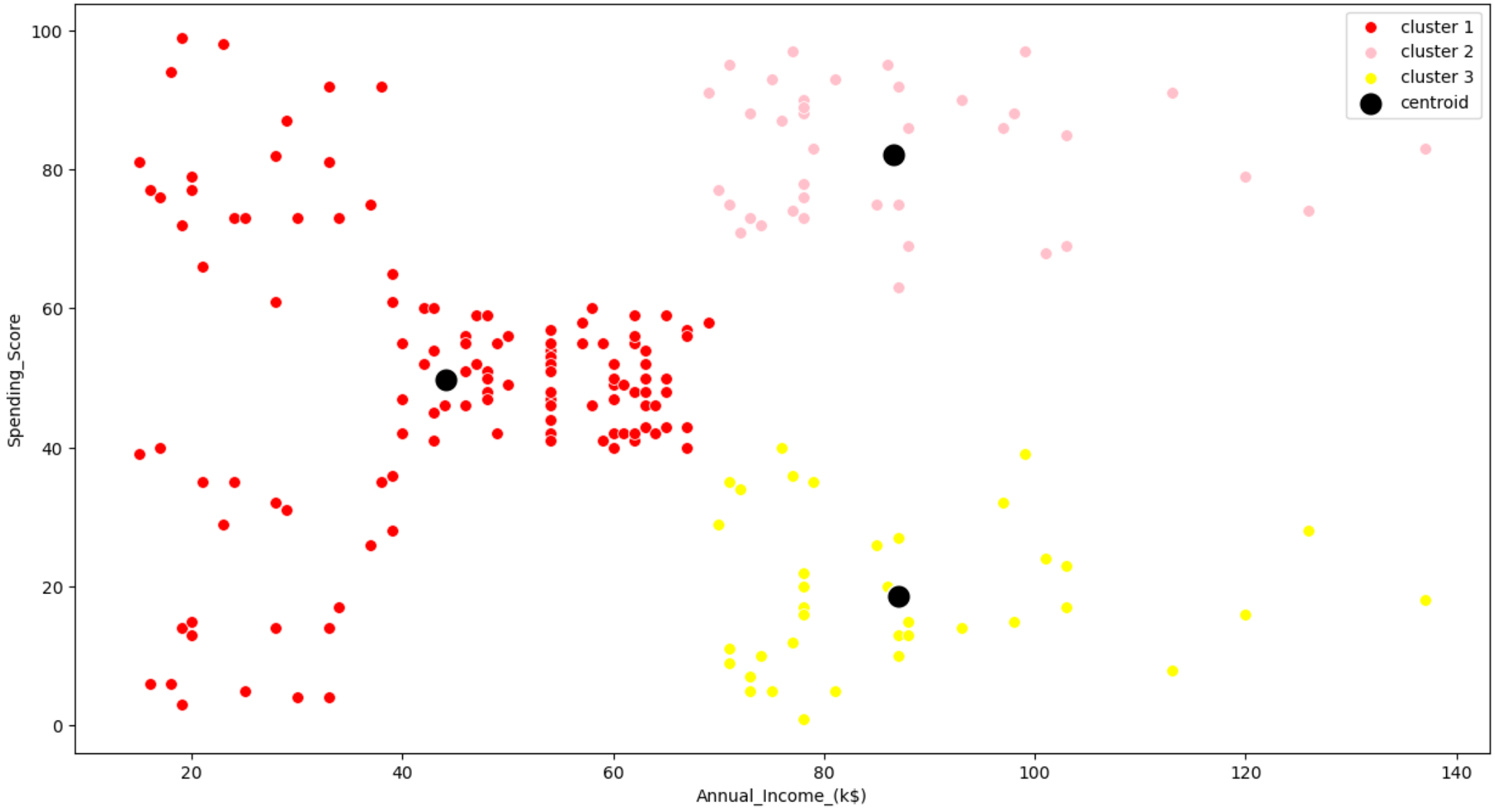
Mall_customers Dataset

CustomerID	Genre	Age	Annual_Income_(k$)	Spending_Score
1	Male	19	15	39
2	Male	21	15	81
3	Female	20	16	6
4	Female	23	16	77
5	Female	31	17	40

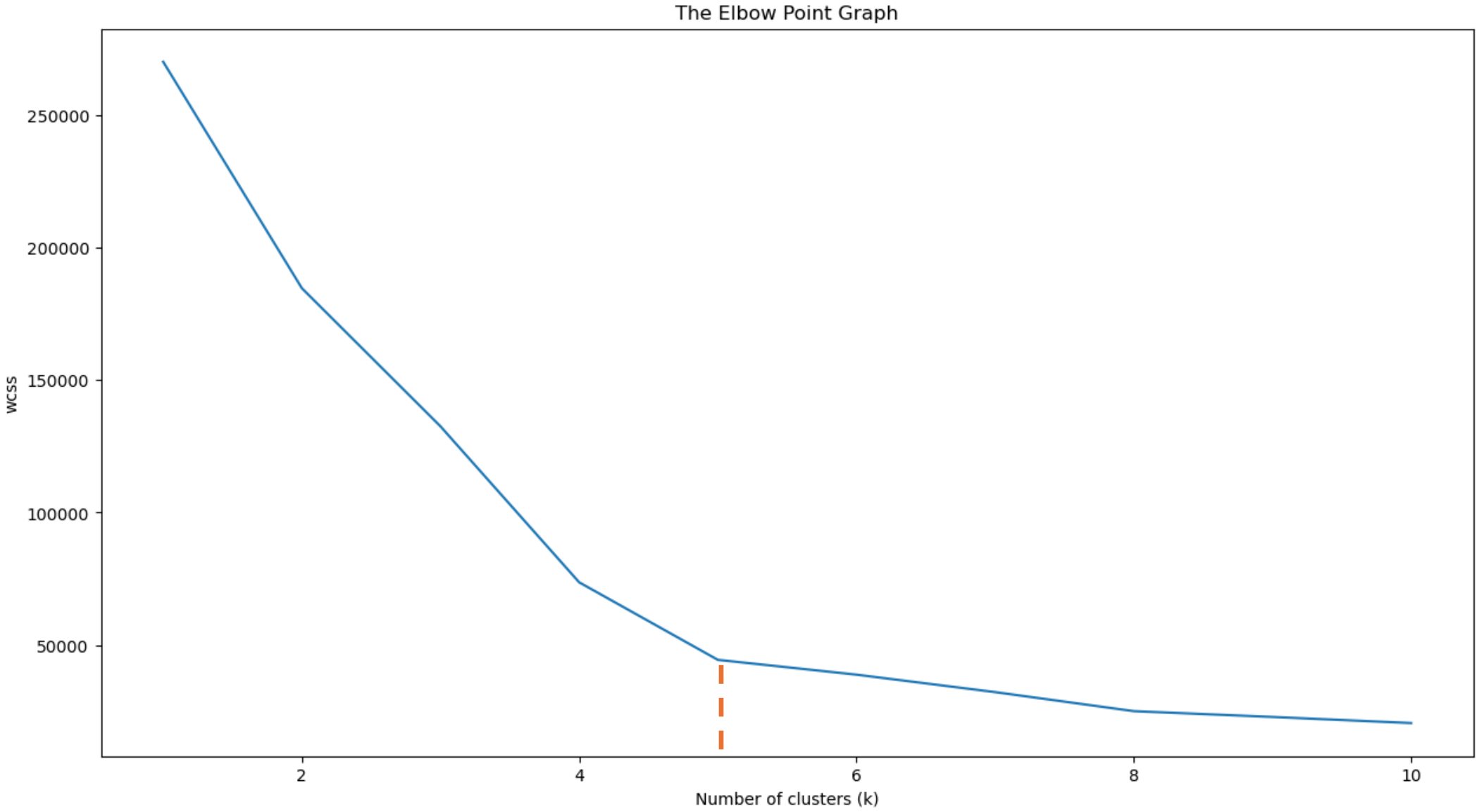


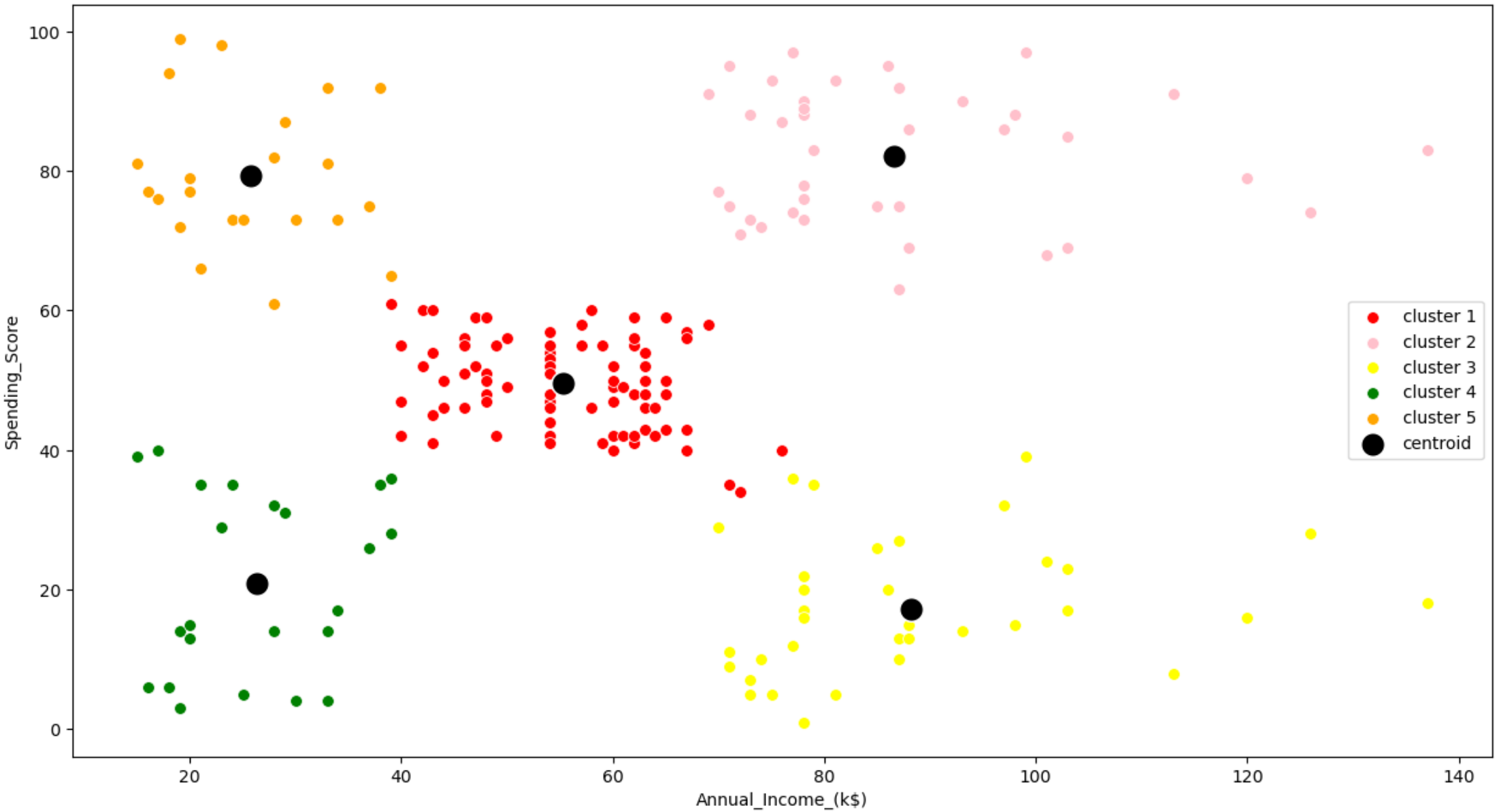
Scatter plot presentation based **Annual_Income_(k$) and Spending_Score**

A **scatter plot** illustrating data points with randomly initialized centroids for k-means clustering with **k=3**



Find the K value using the Elbow method



By using the Elbow method, we can select k value,

The clustering result for k=5,